

STANFORD UNIVERSITY
Department of Electrical Engineering

Lab 3: Switched current sources

Due: Monday, 18 May 2026

Reading: Illustrative function generator schematics; notes on charge-control model (to understand diode and transistor dynamics in the large-signal regime).

Assignment: Design, build, and integrate into your output buffer/hysteretic switch combination with a switched, adjustable current source to generate triangle (and square) waves up to at least 2MHz.

Total frequency coverage: 2Hz to 2MHz, in six bands.

Frequency range per band: $> 11:1$ (you must provide enough overlap so that there are no gaps in frequency coverage in going from one band to the next).

Duty cycle error (square wave): When adjusted to zero at 10kHz, does not exceed 2% at any other frequency *without further adjustment*.

Triangle wave quality: Ratio of maximum amplitude to minimum amplitude should not exceed 1.1 over the entire frequency range. “Blow-by” (feedthrough) should not cause more than 2% shift at peaks.

Supply voltage: $\pm 12\text{V}$ (again, as always)

Allowed components: Same complement as for the output amplifier and hysteretic switch. Op-amps for the current source/sink don’t have to be fancy, so even a 741 should be fine. You will find a JFET all but necessary to buffer the timing capacitor voltage.

Some possibilities: The four-diode gate is a proven, high-performance design. You should feel free to use the commercial example(s) and the *Function Generators* handout as inspiration for your current switch. However, if you wish to explore a transistor-based version, that would be perfectly acceptable as well. There are a great many ways to implement the switch successfully, as has been shown by successful commercial designs.

Important notes: The wafer switches (for switching bands) and big 10k pots (for the main frequency control) are long lead-time items, so please use just one of each per team (that’s all you need).

Finally, note that capacitor tolerances are loose. Trimming through parallel and series combinations may be needed in some bands. You should also consider active capacitance multiplication for the lowest ranges to alleviate that burden. Also, you will need to give some careful thought to what charging current/capacitor value combinations make practical sense (what factors would drive choices one way or the other?).
